



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

SWIFTUI PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Mobile

TOTAL: 10 QUESTIONS

Q1 What is SwiftUI and what UI problem does it address? [Threat Model](#)

Q6 How do you fetch data and handle loading/error states in SwiftUI? [Scalability](#)

Q2 How do components communicate in SwiftUI? [Architecture](#)

Q7 What performance techniques apply to SwiftUI applications? [Lifecycle](#)

Q3 How does SwiftUI manage state for complex applications? [Migration](#)

Q8 How do you handle forms and validation in SwiftUI? [Compliance](#)

Q4 What rendering strategies does SwiftUI support? [Hardening](#)

Q9 How do you test SwiftUI components? [Testing](#)

Q5 How does SwiftUI handle client-side routing? [SSO / IdP](#)

Q10 What accessibility (a11y) practices are essential in SwiftUI? [Training](#)



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 **SwiftUI is a component-based library or framework**

Mitigation

SwiftUI is a component-based library or framework for building interactive user interfaces. It provides a reactive programming model where UI updates automatically when underlying state changes, reducing imperative DOM manipulation.

A6 **Use a data fetching hook or library**

Observability

Use a data fetching hook or library that returns { data, isLoading, error }. Show a skeleton or spinner while loading, an error message on failure, and the data on success. Avoid fetching in render to prevent waterfalls.

A2 **Parent-to-child communication uses props (one-way data flow)**

Initiatives

Parent-to-child communication uses props (one-way data flow). Child-to-parent uses events or callbacks. Siblings share state via a common parent or a global store (Redux, Pinia, Signals). Avoid prop-drilling by using context or state management.

A7 **Code splitting loads only the code needed**

Automation

Code splitting loads only the code needed for the current route. Memoization (useMemo, React.memo) prevents unnecessary re-renders. Virtualization renders only visible list items. Images use lazy loading and next-gen formats (WebP, AVIF).

A3 **Local component state handles UI-only concerns. Shared state**

Rendering

Local component state handles UI-only concerns. Shared state (user auth, cart) lives in a global store with a predictable update pattern (actions, reducers, or mutations). Server state (API data) is managed by libraries like React Query or SWR.

A8 **SwiftUI manages form state with controlled inputs**

Governance

SwiftUI manages form state with controlled inputs or a form library (React Hook Form, VeeValidate). Validation runs on blur or submit, displaying field-level error messages. Schema libraries (Zod, Yup) define validation rules declaratively.

A4 **SwiftUI supports CSR (browser renders), SSR (server renders)**

Gotchas

SwiftUI supports CSR (browser renders), SSR (server renders HTML on each request), SSG (pages generated at build time), and ISR (pages regenerated on a schedule or on-demand). Choose based on SEO requirements and data freshness needs.

A9 **Unit tests check individual component rendering and interactions**

Assessment

Unit tests check individual component rendering and interactions using a component testing library. Integration tests verify multi-component flows. End-to-end tests (Playwright, Cypress) simulate real user journeys in a browser.

A5 **SwiftUI's router (built-in or a library like React Router)**

Federation

SwiftUI's router (built-in or a library like React Router) maps URL paths to components without page reloads. Dynamic segments (/users/:id) are parsed and passed as params. Guards or middleware can protect routes requiring authentication.

A10 **Use semantic HTML (button, nav, main) instead**

Optimization

Use semantic HTML (button, nav, main) instead of divs with click handlers. Ensure keyboard navigability (tab order, focus management). Add ARIA roles and labels for dynamic content. Test with a screen reader and run axe or Lighthouse audits.